

Last time

(Dieudonné - Manin)

- The cat. of isocrystals / $\bar{\mathbb{F}}_p$ is semi-simple w/ simple objs $D(\lambda)$, $\lambda \in \mathbb{Q}$

$$\check{\mathcal{O}}_p := W(\bar{\mathbb{F}}_p)[1/p], \quad \varphi = \text{lift of } \varphi_{\bar{\mathbb{F}}_p}$$

$$M \in \text{Isoc}(\bar{\mathbb{F}}_p) := \check{\mathcal{O}}_p\text{-mod. w/ } \varphi_M : \varphi^* M \xrightarrow{\sim} M$$

$$D(d/n) = \check{\mathcal{O}}_p^n, \quad \varphi_\lambda \leftrightarrow \begin{pmatrix} 0 & & & \\ & \ddots & & \\ & & 0 & \\ & & & \ddots & \\ & & & & 0 \end{pmatrix}$$

- Main goal: $\text{Bun}_{X^{\text{FF}}} \hookrightarrow \text{Isoc}(\bar{\mathbb{F}}_p)$ ③

↑ bij. on objs, NOT equiv of cats

- Harder - Narasimhan formalism applies to $\text{Isoc}(\bar{\mathbb{F}}_p)$ ($r_k = \dim_{\check{\mathcal{O}}_p} D$, $\deg D = -v(\det(\varphi_D))$)

- Classification Theorem: Let $\mathcal{E}$ be a vb on X^{FF} . Then $\exists!$ seq. of rat'l numbers $\lambda_1 \geq \dots \geq \lambda_m$ s.t.

$$\mathcal{E} \simeq \bigoplus_{i=1}^m \mathcal{O}_{X^{\text{FF}}}(\lambda_i).$$

- Functor $\mathcal{E}(-) : \text{Isoc}(\bar{\mathbb{F}}_p) \rightarrow \text{Bun}_{X^{\text{FF}}}$

$$D \mapsto \pi_{n,*}(\mathcal{O}_{X_{E_n}^{\text{FF}}}(d))$$

$$X_{E_n}^{\text{FF}} \simeq X^{\text{FF}} \times_{\mathbb{A}_p} E_n$$

$$\downarrow \pi_n \quad \textcircled{2}$$

$E_n / \mathbb{A}_p$ unr. of deg n

$$X^{\text{FF}} = \text{Proj} \left(\bigoplus_{k \geq 0} B^{\varphi = p^k} \right) \quad X_{E_n}^{\text{FF}} = \text{Proj} \left(\bigoplus_{k \geq 0} B^{\varphi^n = p^k} \right)$$

$$\mathcal{O}(d) := \bigoplus_{k \geq 0} B^{\varphi = p^{k+d}}$$

$$\text{Proj} \left(\bigoplus_{k \geq 0} B^{\varphi^n = p^{kn}} \right) \xrightarrow{\text{Gal}(E_n/\mathbb{A}_p)} X^{\text{FF}}$$

E_n -v.s. $\psi = p^{-n} \varphi$ loc. fin.

$$\text{Hilbert 90} \Rightarrow B^{\varphi^n = p^{kn}} \simeq B^{\varphi = p^k} \otimes E_n$$

$$\text{Div}(X^{\text{FF}}) \xrightarrow{\deg} \mathbb{Z}$$

- First step: $\text{Pic}(X^{\text{FF}}) \simeq \mathbb{Z}$

$$\text{inj. } \forall x \in X^{\text{FF}}, \mathcal{O}(x) \simeq \mathcal{O}(1) \quad \textcircled{1}$$

$$\text{surj. } H^0(X^{\text{FF}}, \mathcal{O}(1)) \xrightarrow{\sim} B^{\varphi = p^n} \quad \textcircled{1}$$

- More gen., $H^i(X^{FF}, \mathcal{O}_X(d)) = \begin{cases} B^{\varphi=P^d}, & i=0 \\ 0 & i=2, \text{ or } i=1 \text{ \& } n \geq 0 \\ B_{dR,X}^+ / \text{Fil}^{-n} B_{dR,X}^+ + \mathbb{Q}_p \neq 0, & i=1 \text{ \& } n \leq -1 \end{cases}$

Today (+ next week)

- ① Compute coh. of line bds on X^{FF}
- ② Study covers of X^{FF} & show that $\pi_1^{\text{ét}}(X^{FF}) \simeq \text{Gal}_{\mathbb{Q}_p}$
- ③ Sketch some key ideas in the proof of the classification theorem
- ④ Weakly adm. $\Rightarrow$ adm.

§ 1. Cohomology of $\mathcal{O}(d)$ on X^{FF}

Assume C^b alg. cl. throughout

Recall: $P := \bigoplus_{k \geq 0} B^{\varphi=P^k}$ is graded factorial w/ irred elts in deg 1.

lem: $H^0(X^{FF}, \mathcal{O}_X(d)) \simeq B^{\varphi=P^d}$

Pf: Since $\mathcal{O}_X(d) = \bigoplus_{k \geq 0} B^{\varphi=P^{k+d}}$, $\exists$ nat'l map

$$B^{\varphi=P^d} \rightarrow H^0(X^{FF}, \mathcal{O}_X(d)) \ni s$$

inj. because P is factorial.

surj. same proof as in $\mathbb{P}^n$ case

(cover X^{FF} by $D_+(t) = ((P)_t)_0$ w/ t irred.; a section s becomes an elt. in $((P)_t)_d$)

□

Prop: $H^i(X^{FF}, \mathcal{O}_X(d))$ is computed as above.

Pf: $i=0 \vee i \geq 2$ follows from the fact that X^{FF} can be covered by two open affines. (Pick 2 non-colinear $t, t' \in B^{\varphi=P}$, then $X^{FF} = \text{Spec}(B_t) \cup \text{Spec}(B_{t'})$. Here, $B_t = ((P)_t)_0$ is a PID & we've seen that $X = \text{Spec}(B_t) \cup \text{Spec}(B_{t'})$ & $X = D_+(t) \cup V_+(t)$, $V_+(t) = \text{Proj}(P.tP) = \{pt\}$)

$i=1$. Consider the SES ($m \geq 0$)

$$0 \rightarrow \mathcal{O}_X(m) \xrightarrow{t} \mathcal{O}_X(m+1) \rightarrow k(\infty_t) \rightarrow 0$$

(this is analogous to $\mathbb{P}^1$; look at the stalks to check this is indeed an SES)

(2)

$$\begin{aligned} \text{LES} \Rightarrow 0 \rightarrow H^0(X^{\text{FF}}, \mathcal{O}_{X^{\text{FF}}(m)}) &\rightarrow H^0(X^{\text{FF}}, \mathcal{O}_{X^{\text{FF}}(m+1)}) \rightarrow k(\infty) \xrightarrow{\delta} \\ &\rightarrow H^1(X^{\text{FF}}, \mathcal{O}_{X^{\text{FF}}(m)}) \rightarrow H^1(X^{\text{FF}}, \mathcal{O}_{X^{\text{FF}}(m+1)}) \rightarrow 0 \rightarrow 0 \dots \end{aligned}$$

Key: $\delta = 0$ ("the fundamental exact sequence in p-adic Hodge theory")

Given this,

$$\begin{aligned} H^1(X^{\text{FF}}, \mathcal{O}_{X^{\text{FF}}}) &\simeq \varprojlim_m H^1(X^{\text{FF}}, \mathcal{O}_{X^{\text{FF}}(m)}) \simeq H^1(X^{\text{FF}}, \varprojlim_m \mathcal{O}_{X^{\text{FF}}(m)}) \\ &\simeq H^1(\text{Spec}(B_t), \mathcal{O}_{\text{Spec}(B_t)}) = 0, \end{aligned}$$

$\simeq j_* \mathcal{O}_X$ where $j: \text{Spec}(B_t) \hookrightarrow X^{\text{FF}}$

Now, it remains to show that the seq. $\downarrow$ until it corresp. to $x \in X^{\text{FF}}$

$$0 \rightarrow B^{\varphi=p^m} \xrightarrow{\cdot t} B^{\varphi=p^{m+1}} \rightarrow B_{dR, X}^+ / \mathfrak{S}_X \simeq K_X \rightarrow 0$$

Note that, $B^{\varphi=p} / t B^{\varphi=1} \rightarrow K_X$

for any $s \in B^{\varphi=p}$ $\downarrow \cdot s^m$ $\nearrow s \downarrow \cdot s^m \Rightarrow$ reduce to the case $m=0$
not colinear w/ $B^{\varphi=p^m} / t B^{\varphi=p^m} \rightarrow K_X$ which is something we've seen:

$$\begin{array}{ccccccc} 0 & \rightarrow & \mathbb{O}_p & \rightarrow & B^{\varphi=p} & \xrightarrow{\theta_x} & K_X \rightarrow 0 \\ & & \uparrow & & \uparrow \log & & \\ 0 & \rightarrow & \varepsilon^{\mathbb{O}_p} & \rightarrow & 1 + \mathfrak{m}_c & \rightarrow & K_X \rightarrow 0 \end{array}$$

(over $E/\mathbb{O}_p$ fin., needs Lubin-Tate thy)

For $m \leq -1$, consider

$$0 \rightarrow \mathcal{O}_X^{\text{FF}}(-1) \xrightarrow{\cdot t} \mathcal{O}_X^{\text{FF}} \rightarrow k(\infty) \rightarrow 0$$

$$\text{LES} \Rightarrow H^1(X^{\text{FF}}, \mathcal{O}_X^{\text{FF}}(-1)) \simeq K_X / \mathbb{O}_p + \text{Induction on } m. \quad \square$$

Rmk: 1) $H^1(X^{\text{FF}}, \mathcal{O}_X(m)) = 0$ for $m \geq 0$ is basically equiv. to the FES.

[Lur] lec 19 established it via a cute coh. computation.

2) X complete alg. curve / alg. cl. field k . Then X has genus 0 $\Leftrightarrow \text{Pic}(X) \simeq \mathbb{Z}$ & $H^1(X, \mathcal{O}_X) = 0$. $\leadsto X^{\text{FF}}$ behaves like a curve of genus 0.

§2. Covers

• $E/\mathbb{O}_p$ fin. extn of deg n

$$\bullet X_E^{\text{FF}} := X^{\text{FF}} \times_{\mathbb{O}_p} E$$

First properties:

③

1) $\pi: X_E^{FF} \rightarrow X^{FF}$ is fin. ét. Since $\text{Spec}(E) \rightarrow \text{Spec}(\mathbb{F}_p)$ is.

2) Since $\mathbb{F}_p \simeq H^0(X, \mathcal{O}_X^{FF})$, $E \simeq H^0(X_E^{FF}, \mathcal{O}_{X_E^{FF}}) \Rightarrow X_E$ is conn.

3) X^{FF} is covered by $\text{Spec}(\mathbb{F}_p[t])$ for $t \in \bigoplus_{k \geq 0} \mathbb{F}_p[t]^k$ homogeneous $\leadsto U_E := U_{X_E^{FF}} E$, which $\simeq U$ can be described by

$$B[t]^{q=1} \otimes_{\mathbb{F}_p} E \simeq (B[t] \otimes_{\mathbb{F}_p} E)^{q=1}$$

covers X_E^{FF}

$\uparrow$ q acts triv. on E

4) $x \in X^{FF}$ cl. pt. then $X_E^{FF} \times_{X^{FF}} \{x\} \simeq \text{Spec}(E \otimes_{\mathbb{F}_p} K_x) = \bigsqcup_n \text{Spec}(K_x) \leadsto$
every cl. pt. of X^{FF} has exactly n pts of X_E lying above it, each having the same res. field.

5) $|X_E^{FF}| \xrightarrow{1-1} \{K, \underbrace{v, u}_{\text{untilt}}\}$, where $u: E \rightarrow K$ is a map of $\mathbb{F}_p$ -alg.

e.g. $E = E_n$ unr. of deg n . then $E = W(\mathbb{F}_p)[t/p]$ &

$$E \rightarrow K \leftrightarrow W(\mathbb{F}_p) \rightarrow \mathcal{O}_K \leftrightarrow \mathbb{F}_p \rightarrow \mathcal{O}_K/p\mathcal{O}_K \leftrightarrow \mathbb{F}_p \rightarrow \mathcal{O}_K^b/\mathfrak{m}$$

n elts cyclically permuted by φ_{C^b}

$$\leftrightarrow \mathbb{F}_p \rightarrow C^b \xrightarrow{\varphi}$$

$$\Rightarrow |X_E^{FF}| \simeq |Y_E|/\varphi^Z \simeq |Y \times \text{Hom}_{\mathbb{F}_p}(\mathbb{F}_p^n, C^b)|/\varphi^Z \simeq |Y|/\varphi^n Z.$$

Thm: $\text{Pic}(X_E^{FF}) \simeq \mathbb{Z}$.

Pf: See [Hur], Lec. 22-25. "Lubin-Tate theory".

□

Thm: X^{FF} is geom. simply conn., i.e., the projection map $X \rightarrow \text{Spec}(\mathbb{F}_p)$ induces an iso. of ét. fund. gps $\pi_1^{\text{ét}}(X^{FF}) \simeq \pi_1^{\text{ét}}(\text{Spec } \mathbb{F}_p) \simeq \text{Gal } \mathbb{F}_p$.

Equiv., pullback along $X^{FF} \downarrow$ induces an equiv. of cats

$$\{ \text{Ét. covers of } \text{Spec}(\mathbb{F}_p) \} \xrightarrow{\sim} \{ \text{---} X^{FF} \}.$$

Lem: $\mathcal{E}, \mathcal{E}'$ vbs on X^{FF} semi-stab. of slope μ, μ' , resp. Then $\mathcal{E} \otimes \mathcal{E}'$ is semi-stab. of slope $\mu + \mu'$.

$\downarrow$ preserve semi-stab. $\downarrow$

$$\text{Pf: WLOG. } \mathcal{E} = \pi_* \mathcal{O}_{X_E^{FF}}(d) \Rightarrow \mathcal{E} \otimes \mathcal{E}' = \pi_* (\mathcal{O}_{X_E}(d) \otimes \pi^* \mathcal{E}').$$

□

Proof of Thm.

Let $\pi: \tilde{X} \rightarrow X$ be a fin. ét. cover. WTS: $\tilde{X} = X \times_{\mathbb{A}^1} E$. $\downarrow$ ét. $\mathbb{A}^1$ -alg
 prod. of fin. extn.
 Put $\mathcal{E} := \pi_* \mathcal{O}_{\tilde{X}}$ & define $E := H^0(X, \mathcal{E})$. ISTS: $\mathcal{E} \simeq E \otimes_{\mathbb{A}^1} \mathcal{O}_X$, i.e.,
 $\mathcal{E}$ is triv. $\xLeftrightarrow[\text{Thm}]{\text{Clsn}}$ $\mathcal{E}$ is semi-stab. of slope 0.

- π fin. ét. $\Rightarrow \mathcal{E} \otimes \mathcal{E} \rightarrow \mathcal{E} \xrightarrow{\text{tr}} \mathcal{O}_X$ non-degen. $\Rightarrow \mathcal{E} \simeq \mathcal{E}^\vee$. But
 $\deg(\mathcal{E}) = -\deg(\mathcal{E}^\vee) \Rightarrow \deg(\mathcal{E}) = 0 \Rightarrow \text{slope } 0 \checkmark$
- Suppose $\mathcal{E}$ is not semi-stab. Let $\mathcal{E}' \subset \mathcal{E}$ be the first step of the HN
 filtn. of $\mathcal{E}$. Then $\mathcal{E}'$ is semi-stab. w/ slope $\mu > 0$ & $\forall \mathcal{F}$ s.s. v.b. of
 slope $> \mu$, $\mathcal{F} \rightarrow \mathcal{E}$ is triv.

$0 = \mathcal{E}_0 \subset \mathcal{E}_1 \subset \dots \subset \mathcal{E}_r = \mathcal{E}$ w/ $\mathcal{E}_i / \mathcal{E}_{i-1}$ s.s. & $\mu_i := \mu(\mathcal{E}_i / \mathcal{E}_{i-1})$ is
 descending strictly. $\downarrow$

lem $\Rightarrow \mathcal{E}' \otimes \mathcal{E}' \hookrightarrow \mathcal{E} \otimes \mathcal{E} \xrightarrow{m} \mathcal{E}$ is triv. But $\exists$ an aff. open $U \subset X$ over
 which $\mathcal{E}'$ has a non-zero section s , & can view s as a non-zero reg. fcn.
 on $\tilde{X} \times_X U$ w/ $s^2 = 0$ $\nexists$.
 $\underbrace{\hspace{1cm}}_{\text{red.}}$ □

$\Rightarrow \{ \text{ét. loc. systems on } X \} \simeq \{ \text{fin. ab gp. w/ Gal}_{\mathbb{A}^1} \text{-action} \}$

Aside: $\pi_1^{\text{ét}}(\text{Spec } \mathbb{A}^1)$

- Choose a geom. pt. $\bar{x} \rightarrow \text{Spec } \mathbb{A}^1 \leftrightarrow$ choosing an alg. cl. $\mathbb{A}^1 \rightarrow \bar{\mathbb{A}}^1$
- Consider the fiber functor $F_{\bar{x}}: \text{Fét}(\text{Spec } \mathbb{A}^1) \rightarrow \text{Sets}$
 $E \mapsto \text{Hom}_{\mathbb{A}^1\text{-Alg}}(E, \bar{\mathbb{A}}^1)$
- $\pi_1^{\text{ét}}(\text{Spec } \mathbb{A}^1) := \text{Aut}(F_{\bar{x}}) \simeq \text{Gal}_{\mathbb{A}^1}$.

§ 3. The Classification Thm.

Criterion (X): For every v.b. $\mathcal{E}$ on X & for every $n \geq 1$, if $\exists S \in \mathcal{S}$

$$0 \rightarrow \mathcal{O}_X(-1/n) \rightarrow \mathcal{E} \rightarrow \mathcal{O}_X(1) \rightarrow 0$$

then $H^0(X, \mathcal{E}) \neq 0$.

cyclic

Thm: Suppose Criterion (X_n) holds for every $\checkmark$ unr. covering X_n of deg n , $\forall n$; then i) The semi-stab vbs on X are isoclinic (pure)

ii) The HN filn always splits

iii) Every vb. on X is iso. to $\bigoplus_{i=1}^m \mathcal{O}_X(\lambda_i)$ for unique rat'l numbers $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_m$.

(Furthermore, if (X_n) holds only for vbs of $rk \leq r$, $\forall n$, then i) - iii) $\checkmark$ for vb. of $rk \leq r$.)

Pf skt: Observe i) $\Rightarrow$ ii) $\Rightarrow$ iii)

$$\text{Ext}^1(\mathcal{O}(\lambda), \mathcal{O}(\mu)) = H^1(X, \mathcal{O}(\mu - \lambda)) = 0 \text{ if } \mu \geq \lambda$$

$$\text{Toy case: } 0 \rightarrow \mathcal{E}_1 \rightarrow \mathcal{E} \rightarrow \mathcal{E}_2 \rightarrow 0$$

$$\mu_1 > \mu_2$$

$\Rightarrow$ splits. $\checkmark$ In gen., induction on length, rk

Step 1. $\mathcal{E}$ is s.s. $\Leftrightarrow \pi^* \mathcal{E}$ is, so wlog pull back to X_n and assume that $\mu(\mathcal{E}) \in \mathbb{Z}$. Twisting by $\mathcal{O}(m)$ remains s.s. $\Rightarrow$ wlog, $\mu(\mathcal{E}) = 0$.
ISTS: If $\mathcal{E}$ is s.s. of slope 0, then $\mathcal{E}$ is triv.

Induction on $rk_{X_n}(\mathcal{E})$. Assume $\checkmark$ for $rk \leq r$.

Step 2. Write $rk(\mathcal{E}) = r+1$, consider $\pi_r^*(\mathcal{E})$. Let $\mathcal{L} \subset \pi_r^*(\mathcal{E})$ be a line bdl of max. deg d . Consider SES

$$0 \rightarrow \mathcal{L} \rightarrow \pi_r^*(\mathcal{E}) \rightarrow \mathcal{E}' \rightarrow 0 \quad (*)$$

$$\Rightarrow d \leq 0 \leq \mu(\mathcal{E}').$$

Step 3. $\mathcal{E}$ is triv. if $d = 0$. Indeed, $\mathcal{L}$ s.s. of slope 0. But the cat. of ss. vb. of slope 0 is ab. $\Rightarrow$ so is $\mathcal{E}'$. Induction on rk . $\Rightarrow \mathcal{E}'$ is triv. $\Rightarrow \mathcal{E}$ is triv. since $\text{Ext}^1(\mathcal{O}^r, \mathcal{O}) = 0$.

Step 4. It is impossible to have $d \leq -2$.

Claim: $\exists \lambda \geq 0$ w/ $\mathcal{O}_{X_r}(\lambda) \hookrightarrow \mathcal{E}'$. (Induction on rk). $d \leq -2 \Rightarrow \exists$ non-triv. hom. $\mathcal{O}_{X_r}(d+2) \rightarrow \mathcal{O}_{X_r}(\lambda) \subset \mathcal{E}'$. Pull back SES $(*) \Rightarrow$ SES

$$0 \rightarrow \mathcal{L} \rightarrow \mathcal{E}'' \rightarrow \mathcal{O}_{X_r}(d+2) \rightarrow 0$$

$\Leftrightarrow$

$$0 \rightarrow \mathcal{L}(-d-1) \rightarrow \mathcal{E}''(-d-1) \rightarrow \mathcal{O}_{X_r}(1) \rightarrow 0$$

$$\stackrel{\text{b}}{\hookrightarrow} \mathcal{O}_{X_n}(-1) \quad (b)$$

Criterion $(X_r) \Rightarrow H^0(X_r, \mathcal{E}''(-d-1)) \neq 0 \Leftrightarrow \exists$ non-trivial hom.

$$\mathcal{O}_{X_r}(d+1) \rightarrow \mathcal{E}''.$$

Since $\mathcal{E}'' \rightarrow \pi_r^*(\mathcal{E})$ is mono., we get a non-trivial $\mathcal{O}_{X_r}(d+1) \xrightarrow{\sim} \pi_r^*(\mathcal{E})$. The line sub-bdl spanned by the image of v has $\deg \geq d+1$. $\#$.

Step 5 It is impossible to have $d = -1$.

See e.g. [Sha], Thm. 7.9

□

Big picture.

Reduction to deg 1 modifications of vbs

C'(X): If $\mathcal{O}_X(1|n)$ is an increasing modn of $\mathcal{E}$ of deg 1, i.e., $\exists$ SES
 $0 \rightarrow \mathcal{E} \rightarrow \mathcal{O}_X(1|n) \rightarrow \mathcal{F} \rightarrow 0$
 $\leftrightarrow$ 1-dim formal h T gp of ht n

w/ $\mathcal{F} = i_{x,*} k(x)$ for a cl. pt. x , then $\mathcal{E} \simeq \mathcal{O}_X^n$.

2) If $\mathcal{E}$ is an increasing modn. of $\mathcal{O}_X^n$ of deg 1, i.e.,

$$0 \rightarrow \mathcal{O}_X^n \rightarrow \mathcal{E} \rightarrow \mathcal{F} \rightarrow 0$$

w " " , then $\mathcal{E} \simeq \mathcal{O}_X^{n-r} \oplus \mathcal{O}_X(1|r)$ for some $r \in \{1, \dots, n\}$.

Thm: $C'(X) \Leftrightarrow C(X)$.

Pf: See e.g. Chao Li's notes "Vector bundles on the Fargues - Fontaine curve".

vb.

A modn of a vb. at $x \leftrightarrow$ a B_{dR}^+ -lattice inside the B_{dR} -v.s. assoc. to the
 $\xrightarrow{\text{deg 1}}$ "minuscule" condn
 $\leadsto$ We have reduced a global problem on the curve to a local linear alg. problem over B_{dR}^+ .

How do we classify all possible B_{dR}^+ -lattices of $\mathcal{O}_X(1|n)$ & $\mathcal{O}_X^n$?

$\Downarrow$
 period lattices of p -div. gps

$\leadsto$ Study period maps on certain Rapoport deformation space of p -div. gps.

- 1) Surj. of the G.-M. period map from the Lubin - Tate sp. to $\mathbb{P}^{n-1}$
- 2) The image of the de Rham period of the RZ sp. of special formal $\mathcal{O}_B$ -mods (Drinfel'd.)

1) $\mathcal{O}_X(1/n) = \mathcal{E}(D, \varphi)$ where $D =$ isocrystal of the unique 1-dim. formal LT gp/H of ht n / $\overline{\mathbb{F}}_p$

min. modn $\leftrightarrow$ specific B_{dR}^+ -lattice in $D \otimes B_{dR}^+$

$\leftrightarrow$ trin Fil D_{K_x} in $D_{K_x} = D \otimes_{\mathbb{Q}_p} K_x$ of dim $n-1$

$\leftrightarrow$ pt in $\mathbb{P}^{n-1}(K_x)$

$\begin{matrix} \text{RZ} \\ \text{defn sp.} \end{matrix} \rightarrow \begin{matrix} \uparrow \pi_{GM} \\ D(\mathcal{O}_{K_x}) \end{matrix}$

Upshot: Every min. modn. comes from a p -div. gp $G/\mathcal{O}_{K_x}$. X^{FF}
of $\mathcal{O}_X(1/n)$

Scholze-Weinstein Thm: $\{p\text{-div. gp. } / \mathcal{O}_{K_x}\} \simeq \{ \text{min. modn. of triv. vbs on } \}$
 $\Rightarrow \mathcal{E}$ is triv. $\underbrace{\hspace{10em}}_{\text{defines a pt. in } \mathbb{P}^{n-1}_{K_x}}$

2) LES in coh. gives $0 \rightarrow H^0(X, \mathcal{E}) \rightarrow \mathbb{Q}_p^n \xrightarrow{ev} K_x \rightarrow \dots$

Claim: $H^0(X, \mathcal{E}) = 0 \Leftrightarrow \mathcal{E} \simeq \mathcal{O}_X(-1/n)$. Indeed, $H^0(X, \mathcal{E}) = 0 \Rightarrow ev$ is inj. $\Leftrightarrow P \in \Omega^{n-1} :=$ open complement of all $\mathbb{Q}_p$ -rat'l hyperplanes in the adic space assoc. to $\mathbb{P}^{n-1}_{K_x}$

Drinfeld: $\mu(K_x) \xrightarrow{\pi_{dR}} \mathbb{P}^{n-1}(K_x)$ has image precisely Ω^{n-1} .

$\uparrow$ certain defn. sp. of p -div. gps w/ additional str.

$\xrightarrow{SW} \mathcal{E} \simeq \mathcal{O}_X(1/n)$.

In general, $H^0(X, \mathcal{E}) \neq 0$ & so the pt lies on at least 1 $\mathbb{Q}_p$ -rat'l hyperplane. Intersect all rat'l hyperplanes containing $P \rightsquigarrow$ get a min. rat'l subspace $V \subset \mathbb{Q}_p^n$ of dim. m .

$\rightsquigarrow P \in \Omega^{m-1} \subset V \otimes K_x = K_x^m$ $V^\perp$ untouched (triv. modn.)

$\downarrow$ $\mathcal{O}_X(-1/m)$ $\downarrow$ $\mathcal{O}_X^{n-m}$